

ENCRYPTION TOOL

Project by :

231210046 OM Pradeep
231210057 Kaushlendra
231210058 Kesav



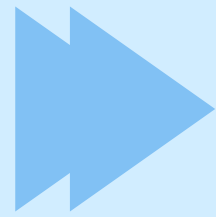
PROBLEM STATEMENT

- Securing sensitive files remains a major challenge for individuals and organizations.
- Files stored or shared in plain format are vulnerable to unauthorized access, data breaches, and cyberattacks.
- Lack of proper encryption mechanisms leads to risks in data confidentiality, integrity, and privacy.
- Without strong encryption, sensitive information such as personal data, financial records, and business documents can be easily compromised.
- Inefficient or complex encryption solutions also create usability barriers, reducing adoption among users.

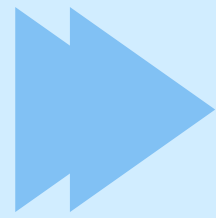




OBJECTIVES



Develop an intuitive file encryption tool that simplifies the process of securing files using strong encryption techniques, making data protection accessible to users of all skill levels.



Enable secure storage and transfer of files by allowing users to easily encrypt and decrypt their data, ensuring confidentiality and preventing unauthorized access across different systems and platforms.

HOW OUR APP SOLVES THE PROBLEM

- Secure File Upload : Simple interface to easily upload files for encryption.
- Strong Encryption (AES + RSA) : Uses AES for fast file encryption and RSA for secure key protection.
- Hybrid Encryption : File encrypted with AES; AES key encrypted with RSA for enhanced security.
- Easy Decryption : Only authorized users with the private key can decrypt files.
- Data Security : Ensures confidentiality and protects data from unauthorized access.
- User-Friendly : Quick and simple process with minimal technical effort.



USER INTERFACE

 **SecureFile**

Military-grade AES & RSA file cryptography

 **Encrypt**

 **Decrypt**

Encryption Method

AES

RSA

 Hybrid asymmetric encryption. Generates a private key file for you automatically.



Drag & Drop your file here

or

Browse Files

Encrypt File →

 **SecureFile**

Military-grade AES & RSA file cryptography

 **Encrypt**

 **Decrypt**

Decryption Method

AES

RSA



Drag & Drop encrypted file (.enc)

Browse Files

 Private Key (.pem)

Select Key

No key selected

Decrypt File 

 **SecureFile**

Military-grade AES & RSA file cryptography

 **Encrypt**

 **Decrypt**

Encryption Method

AES

RSA

 Fast symmetric encryption. Best for large files. Requires a password.



Drag & Drop your file here

or

Browse Files

 Password

Enter a strong password

Encrypt File →

 **SecureFile**

Military-grade AES & RSA file cryptography

 **Encrypt**

 **Decrypt**

Decryption Method

AES

RSA



Drag & Drop encrypted file (.enc)

Browse Files

 Password

Enter your decryption password

Decrypt File 



TECH STACK

- Frontend: HTML, CSS, JavaScript
 - Backend: Python (Flask)
 - Encryption Libraries:
 - AES (via AESGCM)
 - RSA (asymmetric encryption)
 - Security & Key Handling:
 - PBKDF2HMAC (key derivation)
 - hashes module
 - File Handling: os, io, zipfile, werkzeug
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THANK YOU!

